

User Guide – Network Traffic Analysis Tool

1. Purpose of the Tool

This tool analyzes a tcpdump log file in order to detect abnormal activities on the network:

- **Port scans:** systematic sending of SYN packets to consecutive ports
- **Denial of service attacks (DoS):** abnormally high volume of packets sent in a very short time
- **Sensitive SSH connections:** access to the administration server on non-standard ports

The tool converts the raw text file into a CSV file, processes it with Excel/VBA, and produces:

- Filtered sheets for each type of attack
 - Summary charts
 - A report in Markdown format
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2. Prerequisites

Before you begin, verify that you have the following elements:

Software and versions

- **Python 3.x** (version 3.13 tested and validated)
- **Microsoft Excel** recent version supporting macros (.xslm files)
- **Text editor** or **VS Code** to run Python scripts

Python Libraries

The following modules must be installed:

- `pandas` – for data manipulation in Python
- `openpyxl` – to read and write Excel files

Installation (in a terminal):

```
pip install pandas openpyxl
```

Project Structure

The project must be organized as follows:

```
TRAITERLESDONNEES/
├── data/
│   ├── DumpFile.txt (raw tcpdump file)
│   ├── logs_extraits.csv (generated by Python)
│   ├── logs_extraits.xlsm (Excel workbook with macros)
│   ├── rapport_attaques.md (generated report)
│   └── vba.txt (VBA code to copy into Excel)
└── src/
    ├── extractDump.py (CSV extraction script)
    ├── extractICS.py (alternative extraction script)
    └── rapportMd.py (report generation script)
```

3. Step 1 – Generate CSV from tcpdump (Python)

This step converts the raw tcpdump file into a structured CSV file.

Procedure

1. Open a **terminal** (PowerShell or CMD on Windows) in the root folder `TRAITERLESDONNEES`.
2. Verify that the file `DumpFile.txt` is located in the `data/` folder.
3. Run the Python extraction script:
`python src/extractDump.py`
4. A graphical window appears. Click the button "**Choisir un fichier de logs**" and select `DumpFile.txt`.
5. The script analyzes the file and automatically creates `data/logs_extraits.csv` with the following columns:
 - `TIME`: the packet timestamp
 - `SOURCE`: source IP address
 - `DESTINATION`: destination IP address
 - `PORT`: destination port
 - `FLAGS`: TCP flags (SYN, ACK, FIN, etc.)
 - `LENGTH`: packet size in bytes

Verification

The file `logs_extraits.csv` must contain at least several hundred lines. You can open it with Excel or a text editor to verify the format.

4. Step 2 – Analyze data with Excel (VBA Macros)

This step uses a VBA macro to classify packets by attack type and generate charts.

Procedure

1. **Open Excel** and load the file `data/logs_extraits.xlsm`.
2. **Enable macros:** If Excel asks "Do you want to enable this content?", click "**Enable Content**".
3. **Verify the data:** The sheet `logs_extraits` must contain all CSV columns (TIME, SOURCE, DESTINATION, PORT, FLAGS, LENGTH).
4. **Launch the analysis macro:**
 - Go to menu **View → Macros → View Macros...**
 - Select the macro named `Analyse_Reseau_Complet_Clean` (or the final name you kept)
 - Click **Run**
5. **Macro results:** After a few seconds, the macro automatically creates or updates the following sheets:
 - **Onglet_PortScan:** all packets matching a port scan pattern (successive SYN packets)
 - **Onglet_DoS:** all packets matching a DoS attack (high volume in short time)
 - **Onglet_SSH_Sensible:** all packets involving SSH connection on non-standard ports
 - **Synthese:** summary table with total packet count per attack type
 - **Graphiques:** three charts (pie or column charts) showing the distribution of attacks

Exporting Charts

Once the macro is executed, you can export the three charts to integrate them into the Markdown report:

1. In the **Graphiques** sheet, right-click on the first chart → "**Save as Picture...**"
2. Choose the **PNG** format, name the image `graph_nb_attaques.png` and save it in `data/`
3. Repeat for the other two charts and name them:
 - `graph_portscan_ports.png`

- graph_dos_volume.png
4. **Save the workbook:** Ctrl+S (the file remains in .xlsx format).

5. Step 3 – Generate the Markdown report (Python)

This step creates a Markdown file with a summary, example tables, and charts.

Procedure

1. In the **terminal**, make sure you are in the TRAITERLESDONNEES folder.
2. Run the report generation script:
python src/rapportMd.py
3. The script:
 - Reads the sheets Onglet_PortScan, Onglet_DoS, Onglet_SSH_Sensible from the Excel workbook
 - Counts the total number of packets per attack type
 - Creates the file data/rapport_attaques.md with:
 - A global summary of attack counts
 - Tables with 10 example packets for each type
 - References to chart images

Verification

Open data/rapport_attaques.md with a text editor (VS Code, Notepad++, etc.) or view it on GitHub. The file should contain:

- A title "Rapport d'analyse de trafic réseau"
- Three sections: Port scan, DoS, SSH_Sensible
- Three embedded images (with relative paths graph_*.png)
- Readable tables

6. Interpreting the Results

Port Scan

A port scan manifests as:

- A series of **consecutive SYN packets** (flag S) from the same source host

- Towards the same destination host
- With increasing ports (e.g., 2465, 2466, 2467, ..., 2505)
- Over a short time period

Example from the file:

SOURCE: 190-0-175-100.gba.solunet.com.ar

DESTINATION: 184.107.43.74.http

PORTS: 2465 → 2466 → 2467 → ... → 2505

Denial of Service (DoS)

A DoS attack manifests as:

- A **traffic spike**: many packets sent in very short time (same second)
- Towards the same target
- Charts show **visible peaks** at attack times

Example: At 15:34:06, 756 DoS packets were detected in a few seconds, far beyond normal volume.

Sensitive SSH Activity

A sensitive SSH connection manifests as:

- A connection to the server `BP-Linux8.ssh` (or similar)
- On a **non-standard port** (instead of the usual port 22)
- Example: ports 50019, 50245, etc.

Risk: These connections may indicate an unauthorized access attempt or unusual administrative action.

7. Troubleshooting

Problem	Cause	Solution
<code>ModuleNotFoundError: No module named 'pandas'</code>	pandas is not installed	Run <code>pip install pandas</code>
<code>ModuleNotFoundError: No module named 'openpyxl'</code>	openpyxl is not installed	Run <code>pip install openpyxl</code>
Excel macro does not run	Macros are not enabled	Open Excel, click "Enable Content"
File <code>logs_extraits.csv</code> is not created	<code>tcpdump</code> file was not found	Verify that <code>DumpFile.txt</code> exists in <code>data/</code>

Charts do not display in Markdown	PNG images were not exported	Manually export charts from Excel as PNG to <code>data/</code>
File <code>rapport_attaques.md</code> remains empty	Excel sheets do not exist	Verify that sheets <code>Onglet_PortScan</code> , <code>Onglet_DoS</code> , <code>Onglet_SSH_Sensible</code> exist

8. Conclusion

This tool fully automates the analysis of a raw tcpdump file. The three steps (Python → Excel → Python) transform a complex network log into a structured and visual report ready to be shared with the network team or included in documentation.

For any questions or improvements, consult the comments in the Python code or the VBA macros.